KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5093A | 5 Credits: 4 | Program Elective | Source-grounded

Injection Mould Design

 To acquaint with various metals and operations carried out in plastic injection

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5093A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5093A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5093A
Course title	Injection Mould Design
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To acquaint with various metals and operations carried out in plastic injection
- mould making.
- To assimilate design and working principles of various components of plastic
- injection moulds.

Course outcomes

CO	Official outcome	Study level
CO1	15 Applying operations for injection mould making Explain various parts of an injection mould and	See official syllabus
CO2	15 Understanding operations involved in their construction Describe injection mould feed system and ejection	See official syllabus
CO3	14 Applying system	See official syllabus
CO4	Generalize parting surface and cooling systems of 14 Applying injection mould	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 3
CO2	CO2 3 3
CO3	CO3 3 3
CO4	CO4 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	making	4	As specified
Module 2	their construction	4	As specified
Module 3	Describe injection mould feed system and ejection system	4	As specified
Module 4	Generalize parting surface and cooling systems of injection mould	4	As specified

MODULE 1

making

This module is presented from the official syllabus scope for Injection Mould Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: making
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

2 Remembering steel

2 Remembering steel is an official topic in Module 1 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering steel using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering steel should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering steel means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 2 Remembering steel [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their

Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their is an official topic in Module 1 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize injection mould making steel 2 understanding illustrate important machine tools and their should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

9 Understanding operations used in Injection mould making Outline machining operations required for a

9 Understanding operations used in Injection mould making Outline machining operations required for a is an official topic in Module 1 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 9 Understanding operations used in Injection mould making Outline machining operations required for a using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 9 understanding operations used in injection mould making outline machining operations required for a should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 9 Understanding operations used in Injection mould making Outline machining operations required for a means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 1-□□ 9 Understanding operations used in Injection mould making Outline machining operations required for a □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in

2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in is an official topic in Module 1 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying mould plate preparation classification of metals – ferrous and non-ferrous. composition of plain carbon steel. alloy steel – mould steel-types – requirements. list important machine tools used for injection mould making. identify parts of a lathe using line diagram. lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. mode of operation of planer and shaper. comparison between planer and shaper. cylindrical grinding machine – sketch and mode of operation. basic machine tool operations for manufacturing a simple mould plate. explain various parts of an injection mould and operations involved in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 1-□□ 2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
making
            Summarize metals and alloys with main focus on
  M1.01                                                         2
Remembering
            steel
  M1.02     Familiarize injection mould making steel            2
Understanding
            Illustrate important machine tools and their
  M1.03                                                         9
Understanding
            operations used in Injection mould making
            Outline machining operations required for a
  M1.04                                                         2
Applying
            mould plate preparation

Contents:
Classification of metals – ferrous and non-ferrous. Composition of plain carbon
steel. Alloy
steel – Mould steel-types – requirements.
List important machine tools used for injection mould making. Identify parts of a
Lathe
using line diagram. Lathe operations in brief; turning (different types), facing,
parting off,
forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation
of planer and
shaper. Comparison between planer and shaper. Cylindrical grinding machine –
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

their construction

This module is presented from the official syllabus scope for Injection Mould Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: their construction
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Remembering injection mould general terms

2 Remembering injection mould general terms is an official topic in Module 2 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering injection mould general terms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering injection mould general terms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering injection mould general terms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 2 Remembering injection mould general terms [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection

Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection is an official topic in Module 2 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate core and cavity making techniques 6 understanding familiarize different parts of an ideal injection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of

6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of is an official topic in Module 2 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding mould, ancillary parts outline steps involved in bench fitting of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types. Steps involved in Bench fitting of an Injection mould.

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types. Steps involved in Bench fitting of an Injection mould. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding injection mould. understand the terms; injection mould clamping unit - fixed & moving halves, tie bar & bush. terms relating injection moulds - impression, cavity and core. cavity and core making techniques; casting, electro - deposition, cold hobbing, pressure casting. integer and insert type mould. bolsters – requirements and classification. types of cavity and core inserts. sketch of an ideal injection mould and label its parts. ancillary items; guide bush &guide pillar – functions, types, location and positioning. sprue bush – function and types. register ring – function and types. steps involved in bench fitting of an injection mould. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types. Steps involved in Bench fitting of an Injection mould. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 2-□□ 1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types.

Steps involved in Bench fitting of an Injection mould. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
their construction
            Know the injection mould clamping unit and
  M2.01                                                         2
Remembering
            injection mould general terms
  M2.02     Illustrate core and cavity making techniques        6
Understanding
            Familiarize different parts of an ideal injection
  M2.03                                                         6
Understanding
            mould, ancillary parts
            Outline steps involved in Bench fitting of
  M2.04                                                         1
Understanding
            Injection mould.
            Series Test – I                                     1

Contents:
Understand the terms; Injection mould clamping unit - fixed & moving halves, tie
bar &
bush. Terms relating Injection moulds - impression, cavity and core.
Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing,
Pressure
casting. Integer and insert type mould. Bolsters – requirements and
classification. Types of
cavity and core inserts.
Sketch of an ideal injection mould and label its parts. Ancillary items; Guide
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Describe injection mould feed system and ejection system

This module is presented from the official syllabus scope for Injection Mould Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Describe injection mould feed system and ejection system
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and

Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and is an official topic in Module 3 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify injection mould feed system 2 remembering illustrate runner and gate in terms of types and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

6 Understanding designs Explain different parts of an Ejector grid and

6 Understanding designs Explain different parts of an Ejector grid and is an official topic in Module 3 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding designs Explain different parts of an Ejector grid and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding designs explain different parts of an ejector grid and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding designs Explain different parts of an Ejector grid and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 6 Understanding designs Explain different parts of an Ejector grid and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return

3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return is an official topic in Module 3 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding ejector plate assembly illustrate the ejector plate actuation-return should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B.

3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B. is an official topic in Module 3 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding mechanisms and ejection techniques define feed system – sprue, runner and gate. runner cross-sectional shapes and efficiency. runner size and layouts. gates – need for small cross-sectional area, position of gate, balanced gating, types of gates. ejection – ejector grid and designs, ejector plate assembly - ejector plate & retaining plate, ejector plate actuation methods, guiding and supporting of ejector plate assembly, ejector rod and ejector rod bush – designs. ejector plate assembly return systems, stop pins. ejection techniques – different types, ejection from fixed half. sprue pullers – type a & b. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□

□□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Describe injection mould feed system and ejection system

  M3.01     Identify Injection mould feed system                    2
Remembering
            Illustrate runner and gate in terms of types and
  M3.02                                                             6
Understanding
            designs
            Explain different parts of an Ejector grid and
  M3.03                                                             3
Understanding
            Ejector plate assembly
            Illustrate the ejector plate actuation-return
  M3.04                                                             3
Understanding
            mechanisms and ejection techniques

Contents:
Define feed system – sprue, runner and gate. Runner cross-sectional shapes and
efficiency.
Runner size and layouts. Gates – Need for small cross-sectional area, Position of
gate,
Balanced gating, Types of gates.
Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate &
Retaining plate,
Ejector plate actuation methods, Guiding and supporting of ejector plate
assembly, Ejector
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Generalize parting surface and cooling systems of injection mould

This module is presented from the official syllabus scope for Injection Mould Design. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Generalize parting surface and cooling systems of injection mould
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface

Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface is an official topic in Module 4 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate various parting surface and relief 2 understanding explain the significance of mould surface should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding balancing and venting Explain various cooling systems adopted in

2 Understanding balancing and venting Explain various cooling systems adopted in is an official topic in Module 4 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding balancing and venting Explain various cooling systems adopted in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding balancing and venting explain various cooling systems adopted in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding balancing and venting Explain various cooling systems adopted in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding balancing and venting Explain various cooling systems adopted in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

8 Understanding Injection moulds Explain the water connections and seals used in

8 Understanding Injection moulds Explain the water connections and seals used in is an official topic in Module 4 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Understanding Injection moulds Explain the water connections and seals used in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 understanding injection moulds explain the water connections and seals used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Understanding Injection moulds Explain the water connections and seals used in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 8 Understanding Injection moulds Explain the water connections and seals used in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com

2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com is an official topic in Module 4 of Injection Mould Design. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding mould cooling parting surface – classification, non-flat parting surfaces – stepped, profiled, angled & complex edge forms. balancing of mould surfaces, relief of parting surfaces, mould venting. mould cooling - factors deciding mould temperature. types of cooling systems – integer mould plates, integer core plates, insert-bolster assembly, cooling other mould parts – stripper plate, valve type ejector head, sprue bush. water connections and seals in brief –o-ring seals. text / reference: t/r book title/author elements of workshop technology – vol. i & ii – s k hajrah choudhury (media t1 promoters and publishers pvt. ltd.) t2 injection mould design- r.g.w.pye (publisher – john wiley & sons inc.) t3 injection molding handbook - dominick v. rosato(publisher – springer) how to make injection molds - georg menges (publisher - oxford university r1 press, usa) handbook of thermoplastics injection mould design – p s cracknell(publisher - r2 blackie academic & professional) r3 injection mold design engineering - david kazmer (publisher – hanser) online resources: sl.no website link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief – O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Generalize parting surface and cooling systems of injection mould

  M4.01     Illustrate various parting surface and relief           2
Understanding
            Explain the significance of mould surface
  M4.02                                                             2
Understanding
            balancing and venting
             Explain various cooling systems adopted in
  M4.03                                                           8
Understanding
             Injection moulds
             Explain the water connections and seals used in
  M4.04                                                           2
Understanding
             mould cooling
             Series Test – II                                     1

Contents:
Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled,
Angled &
Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces,
Mould
venting.
Mould cooling - Factors deciding mould temperature. Types of cooling systems –
Integer
mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 2 Remembering steel. (3 marks)

Answer: Begin with the standard definition or identity of *2 Remembering steel*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their. (3 marks)

Answer: Begin with the standard definition or identity of *Familiarize injection mould making steel 2 Understanding Illustrate important machine tools and their*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 9 Understanding operations used in Injection mould making Outline machining operations required for a. (3 marks)

Answer: Begin with the standard definition or identity of *9 Understanding operations used in Injection mould making Outline machining operations required for a*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying mould plate preparation Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate. Explain various parts of an injection mould and operations involved in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Remembering injection mould general terms. (3 marks)

Answer: Begin with the standard definition or identity of *2 Remembering injection mould general terms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate core and cavity making techniques 6 Understanding Familiarize different parts of an ideal injection*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding mould, ancillary parts Outline steps involved in Bench fitting of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types. Steps involved in Bench fitting of an Injection mould.. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding Injection mould. Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core. Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts. Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush &Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types. Steps involved in Bench fitting of an Injection mould.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and. (3 marks)

Answer: Begin with the standard definition or identity of *Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding designs Explain different parts of an Ejector grid and. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding designs Explain different parts of an Ejector grid and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding Ejector plate assembly Illustrate the ejector plate actuation-return*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B.. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding mechanisms and ejection techniques Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates. Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding balancing and venting Explain various cooling systems adopted in. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding balancing and venting Explain various cooling systems adopted in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 8 Understanding Injection moulds Explain the water connections and seals used in. (3 marks)

Answer: Begin with the standard definition or identity of *8 Understanding Injection moulds Explain the water connections and seals used in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook

of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding mould cooling Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals. Text / Reference: T/R Book Title/Author Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media T1 Promoters and Publishers Pvt. Ltd.) T2 Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.) T3 Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer) How to make injection molds - Georg Menges (Publisher - Oxford University R1 Press, USA) Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - R2 Blackie Academic & Professional) R3 Injection Mold Design Engineering - David Kazmer (Publisher – Hanser) Online Resources: Sl.No Website Link 1 https://www.dme.net 2 https://techcenter.lanxess.com 3 https://www.rtpcompany.com*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Remembering steel**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Remembering injection mould general terms**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Identify Injection mould feed system 2 Remembering Illustrate runner and gate in terms of types and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Illustrate various parting surface and relief 2 Understanding Explain the significance of mould surface**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.dme.net https://techcenter.lanxess.com https://www.rtpcompany.com

IN-PAGE SEARCH

Search this lesson